

DATA SCIENCE · PHASE 4

Predicting Student Dropout and Academic Success

A Multi-Class Classification Study Using Advanced Supervised Machine Learning

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4,424

Students

39

Features

0.879

Final Model AUC

Why This Matters

- Student dropout costs institutions revenue and harms student futures
- Traditional monitoring reacts too late — after the student has mentally left
- A predictive model on Day-1 enrolment data can flag at-risk students from the start
- Dataset: 4,424 students · Portuguese polytechnic · 39 features · zero missing values

Demographics

5 features

Academic History

4 features

Socio-Economic

8 features

Semester Perf.

12 features

Macroeconomic

3 features

Engineered

3 features

Three Outcome Classes

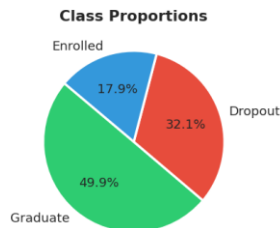
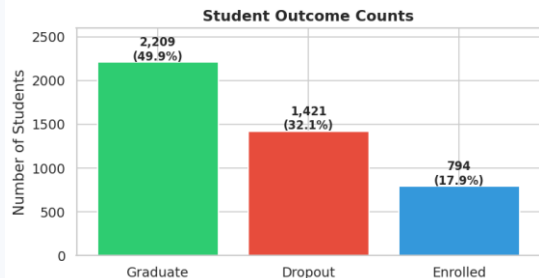
Graduate
49.9% · 2,209

Dropout
32.1% · 1,421

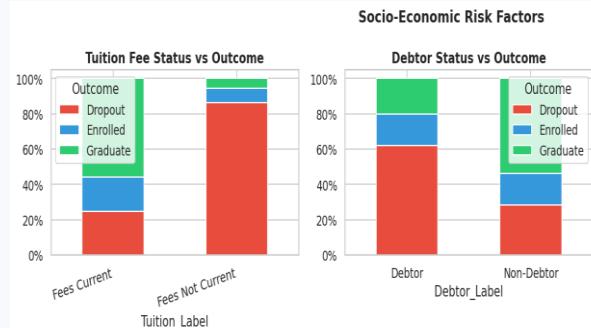
Enrolled
17.9% · 794

Target Variable Distribution

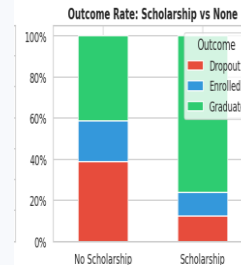
Target Variable: Student Academic Outcome



Tuition Fees & Debtor Status vs Outcome



Scholarship vs Outcome



50/32/18 Class Split — Why Accuracy Fails

Graduate (49.9%) majority means predicting Graduate always gives ~50% accuracy — completely useless. Macro F1 weights all three classes equally and is the sole primary metric throughout.

Tuition Fees: Near-Complete Outcome Reversal

Students with overdue fees drop out at ~75% vs ~10% for fee-current students. SHAP will confirm this as the #2 dropout driver (mean $|SHAP| = 0.761$). First missed payment triggers immediate outreach.

Scholarship Holders Graduate at Dramatically Higher Rates

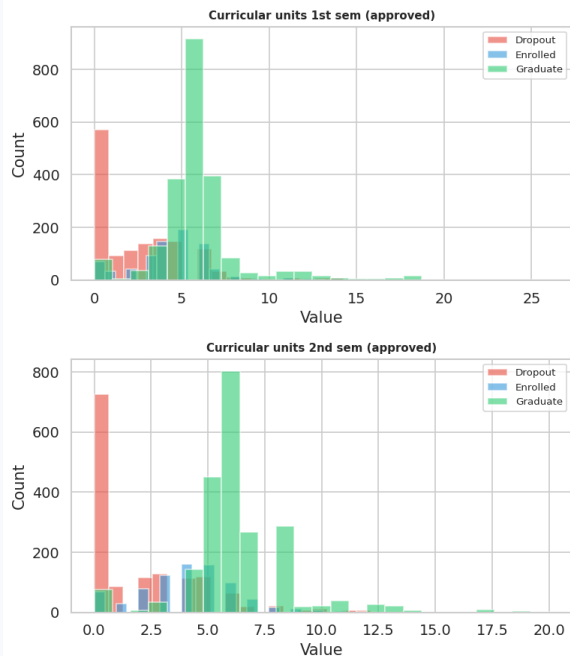
Financial stability is a proven retention mechanism. Scholarship holders show graduation rates roughly 3× higher than non-holders. Expanding coverage is a policy lever, not just a reward.

Strongest Predictors — Semester Academic Performance

04

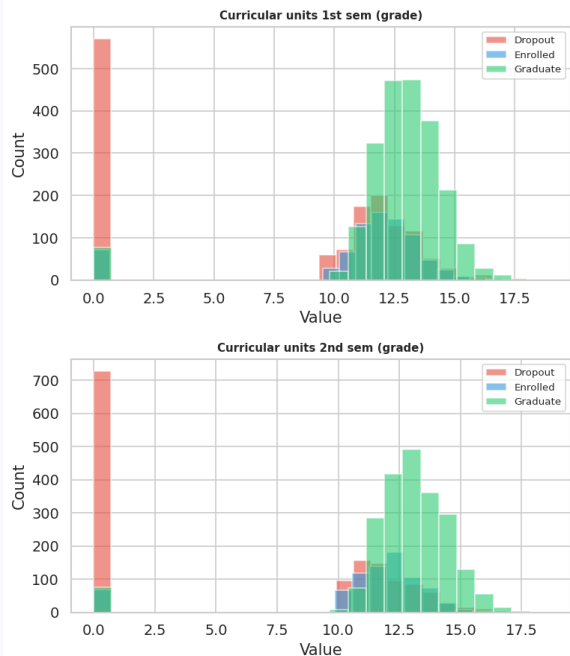
Units Approved — Semester 1 & 2

Semester Performance Distributions by Outcome

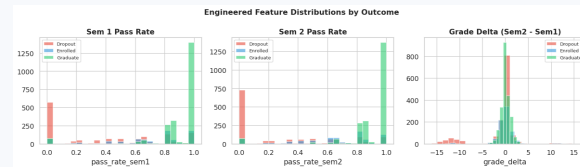


Grades — Semester 1 & 2

Distributions by Outcome



Engineered Trajectory Features



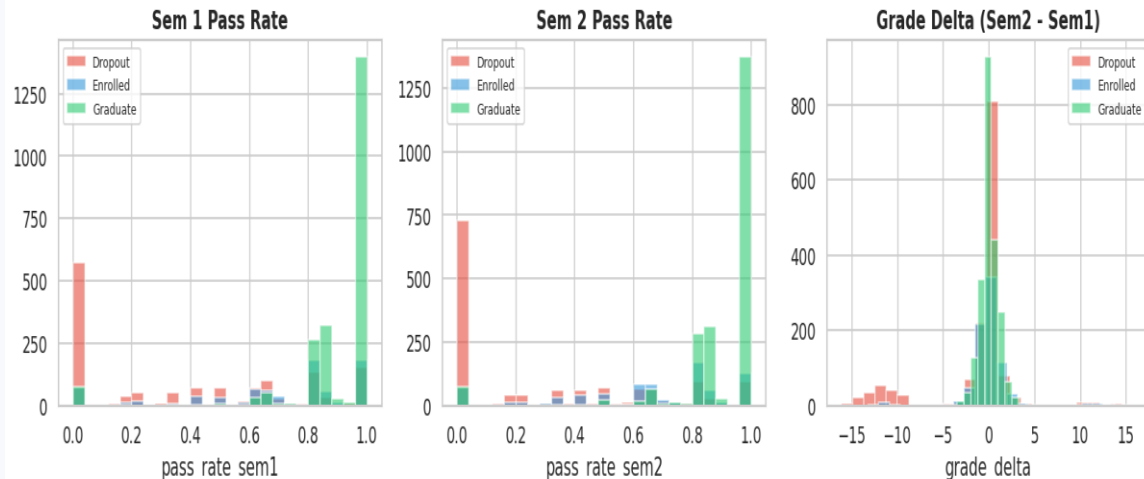
pass_rate_sem2 — #1 SHAP Driver (1.045)

Dropout students cluster at 0 approved units both semesters. A student passing zero units in Sem 1 is nearly always Dropout. Even one unit passed substantially reduces risk.

grade_delta — Trajectory Over Level

grade_delta = Grade Sem2 – Grade Sem1. Positive = improving = graduation trajectory. Negative = declining = dropout trajectory. The SHAP dependence plot shows the risk boundary crosses exactly at delta = 0.

Engineered Feature Distributions by Outcome



All three features are computed before the train/test split and scaled inside cross-validation folds — guaranteed no data leakage.

pass_rate_sem1

$\text{Approved} \div \text{Enrolled (Semester 1)}$

Normalises unit counts — a student passing 4/4 is fundamentally different from one passing 4/8. Rate captures engagement better than raw count.

pass_rate_sem2

$\text{Approved} \div \text{Enrolled (Semester 2)}$

Even stronger separation by Semester 2. If a student has not improved their pass rate, the model's dropout signal intensifies markedly.

grade_delta

$\text{Grade Sem 2} - \text{Grade Sem 1}$

Captures whether the student is on an upward or downward academic trajectory — SHAP shows this feature crosses zero exactly at the risk boundary.

Modelling Strategy — Five Iterations

Each model is chosen to answer one specific diagnostic question about the data's structure

1

Dummy Classifier

F1 = 0.222

What does doing nothing look like?

Floor: predicts Graduate for every student. Zero dropout recall. All subsequent models are judged by how much they close the gap to 1.0

2

Logistic Regression

F1 = 0.685

Is there linear signal in the features?

+0.463 vs baseline in one step — confirms the dataset has strong, linearly separable signal. Most predictive value is linear.

3

KNN (k = 51)

F1 = 0.607

Is the feature space cluster-structured?

Lower than LR — confirms outcomes are NOT cluster-structured. Decision boundaries are complex and non-local. Trees are better suited.

4

Random Forest

F1 = 0.708

Do tree ensembles capture interactions?

+0.023 over LR. Non-linear interactions (e.g. fees overdue AND zero approvals) are captured. Wider train/val gap signals variance.

5

XGBoost

F1 = 0.702

Does sequential error-correction help?

Tighter train/val gap than RF — better regularised. Matches RF at default hyperparameters; needs tuning to reveal its full advantage.



LightGBM (Tuned)

F1 = 0.695

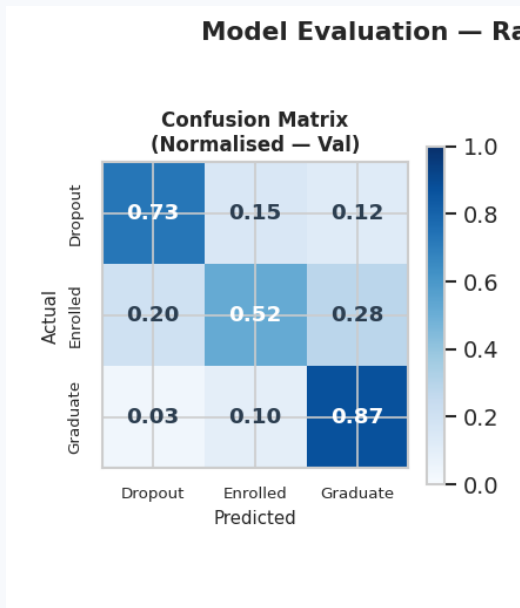
What does the optimised model achieve?

Best calibration + AUC 0.879. RandomizedSearchCV over 12 parameter combinations × 5 CV folds. Final deployed model.

Model 3 — Random Forest · Test Macro F1: 0.708 · Macro AUC: 0.887

07

Normalised Confusion Matrix — What the model gets right and wrong per class

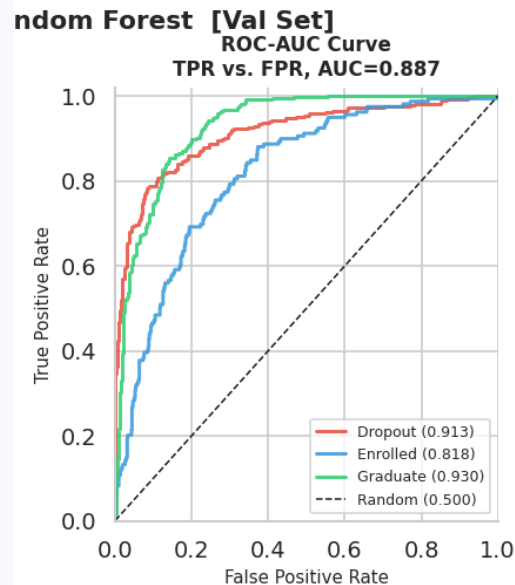


Graduate
87% recall

Dropout
73% recall

Enrolled
52% recall

ROC-AUC Curves — Discriminative power at every possible threshold



Dropout
AUC 0.913

Enrolled
AUC 0.818

Graduate
AUC 0.930

What this tells us:

Graduate 0.87 and Dropout 0.73: compound rules (fees overdue AND zero approvals) are now captured by tree splits. Enrolled 0.52: some non-linear boundaries approximated but misclassification persists between all three classes.

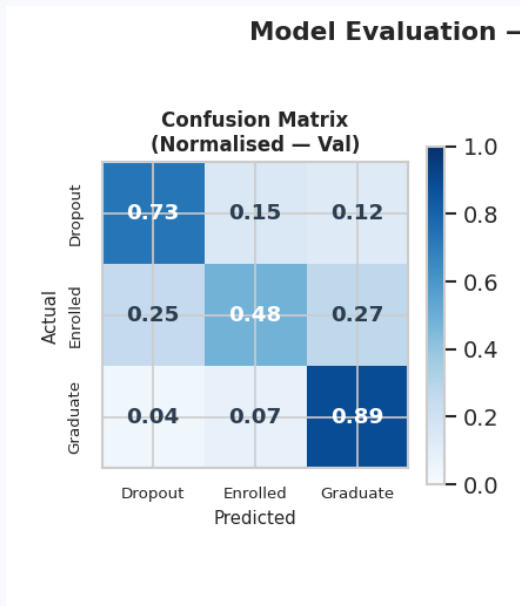
What this tells us:

All three classes well above the 0.50 random baseline. Graduate (0.930) and Dropout (0.913) are strong — the model ranks at-risk students reliably. Enrolled AUC 0.818 is meaningful despite lower classification accuracy.

Model 4 — XGBoost · Test Macro F1: 0.702 · Macro AUC: 0.890

08

Normalised Confusion Matrix — What the model gets right and wrong per class

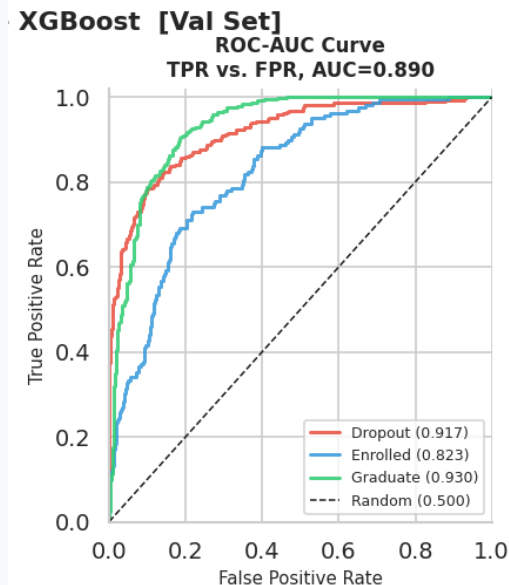


Graduate
89% recall

Dropout
73% recall

Enrolled
48% recall

ROC-AUC Curves — Discriminative power at every possible threshold



Dropout
AUC 0.917

Enrolled
AUC 0.823

Graduate
AUC 0.930

What this tells us:

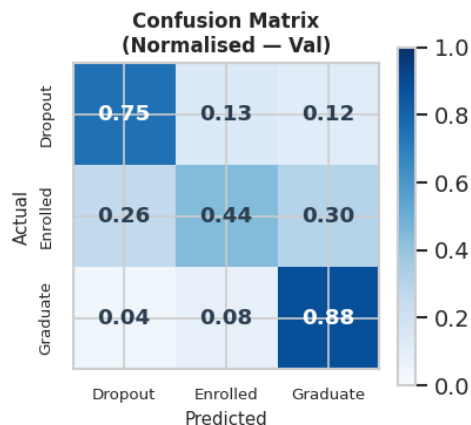
Graduate 0.89 slightly improved. Dropout 0.73 comparable to RF. Enrolled 0.48: sequential error-correction is starting to address the ambiguous middle class. Critical: Dropout→Graduate cell ≈ 0.12 — no false reassurance of at-risk students.

What this tells us:

Enrolled AUC 0.823 improves most vs RF — sequential boosting concentrates effort on hard cases. Graduate 0.930, Dropout 0.917 are strong. Default hyperparameters limit XGBoost; RandomizedSearchCV on LightGBM unlocks the full advantage.

Normalised Confusion Matrix — What the model gets right and wrong per class

Model Evaluation — LightGBM (



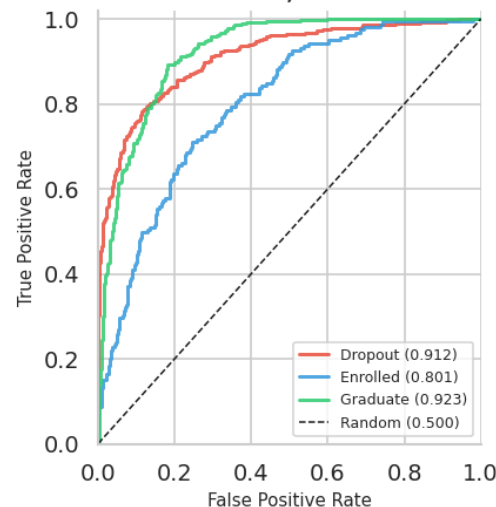
Graduate
88% recall

Dropout
75% recall

Enrolled
44% recall

ROC-AUC Curves — Discriminative power at every possible threshold

Tuned) — Final Model [Val Set]
ROC-AUC Curve
TPR vs. FPR, AUC=0.879



Dropout
AUC 0.912

Enrolled
AUC 0.801

Graduate
AUC 0.923

What this tells us:

Best dropout recall of all models at 0.75. Graduate 0.88. Most important: Dropout→Graduate cell = 0.12 — model virtually never falsely reassures an at-risk student. All Dropout errors redirect to Enrolled (uncertain), never to Graduate.

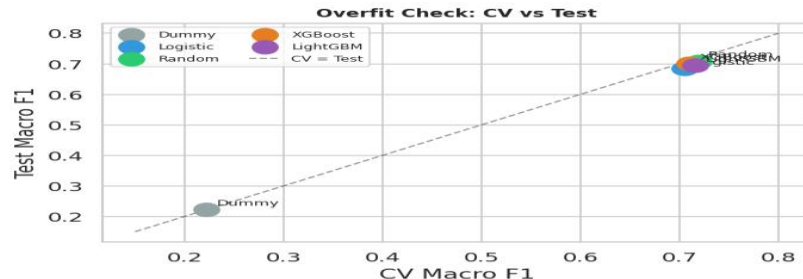
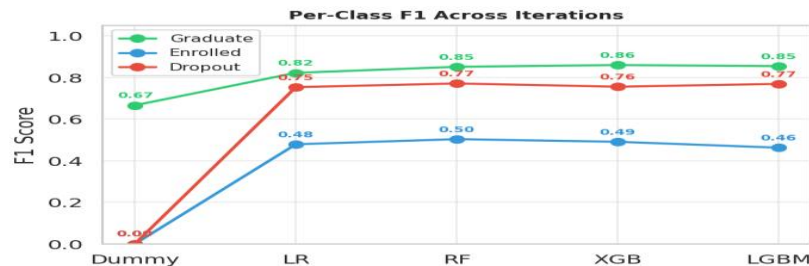
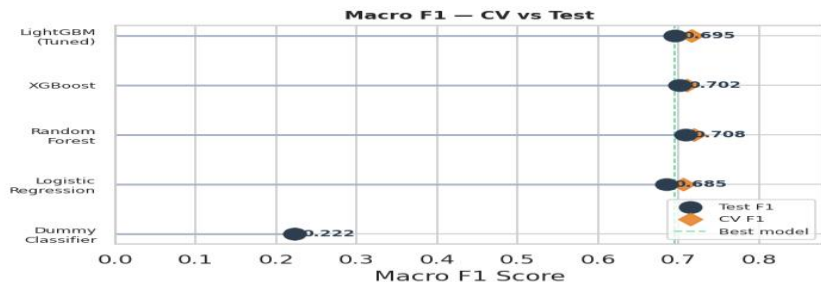
What this tells us:

Macro AUC 0.879. If you draw one Dropout and one non-Dropout randomly, the model ranks the Dropout higher 91.2% of the time — at any threshold. Graduate 0.923 is near-perfect. Enrolled 0.801 is still above chance despite classification difficulty.

Cross-Model Performance Comparison

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Model Performance Dashboard — All Five Iterations



Lollipop: Macro F1 Arc

The largest single jump is Dummy→LR (+0.463). Most signal is linear. RF adds +0.023, LGBM +0.01 more. The law of diminishing returns is visible —

Per-Class: Three Stories

Graduate solved by LR (~0.82) and plateaus at 0.85. Dropout improves through every model. Enrolled never exceeds 0.50 — a structural data

Heatmap: Difficulty Tiers

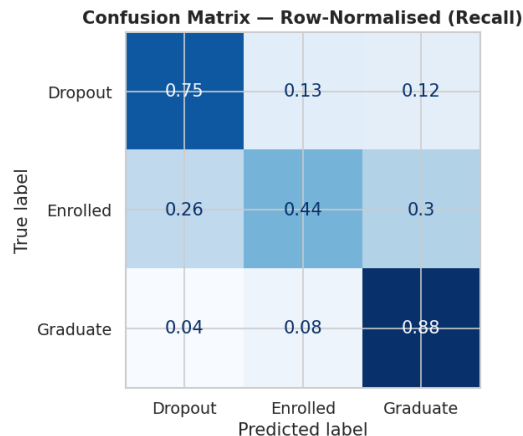
Colour immediately shows three tiers: Graduate = dark green throughout. Dropout = lightens with better models. Enrolled = palest across all five — the persistently ambiguous class.

Scatter: No Leakage

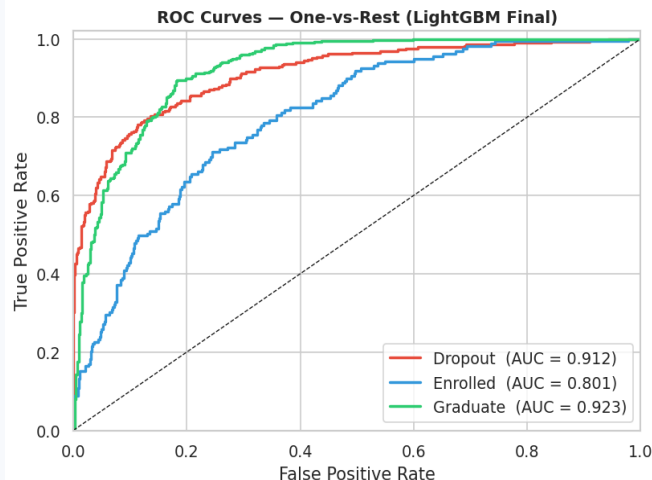
All five models sit on or above the CV=Test diagonal — no model overfit to the training data. The stratified split and SMOTE-inside-CV-only design successfully prevented leakage.

Row-Normalised Confusion Matrix

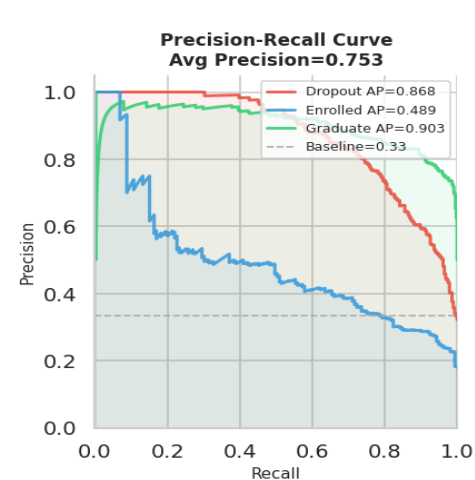
Model Confusion Matrices



ROC-AUC Curves (One-vs-Rest)



Precision-Recall Curves



Confusion Matrix — Per-Class Recall

Graduate 88%: errors to Enrolled, never Dropout.
Dropout 75%: 3-in-4 caught. Dropout→Graduate = 0.12 — near-zero — model never falsely reassures at-risk students. Enrolled 44%: structural ambiguity.

ROC-AUC — Ranking Quality at All Thresholds

Macro AUC 0.879. Dropout 0.912: correctly ranks at-risk vs safe student 91.2% of the time. Graduate 0.923. Enrolled 0.801 — above chance despite hard-label difficulty.

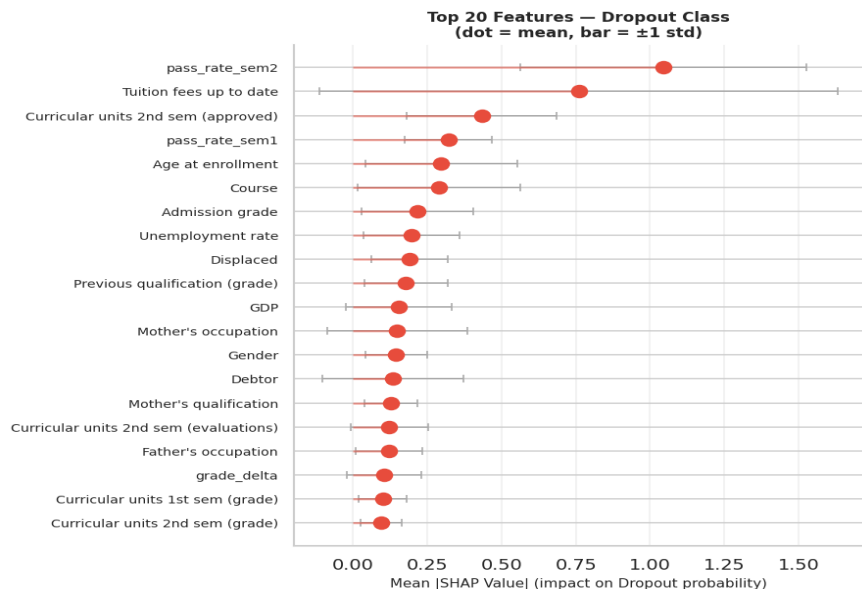
PR Curve — Precision-Recall Trade-off

Dropout AP = 0.868. At 75% Dropout recall, ~85% of flagged students are genuine at-risk. Allows institution to choose operating point based on counsellor capacity.

SHAP Interpretability — What Drives Dropout Risk Globally

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Top 20 Dropout drivers — dot = mean |SHAP|, bar = ± 1 std deviation



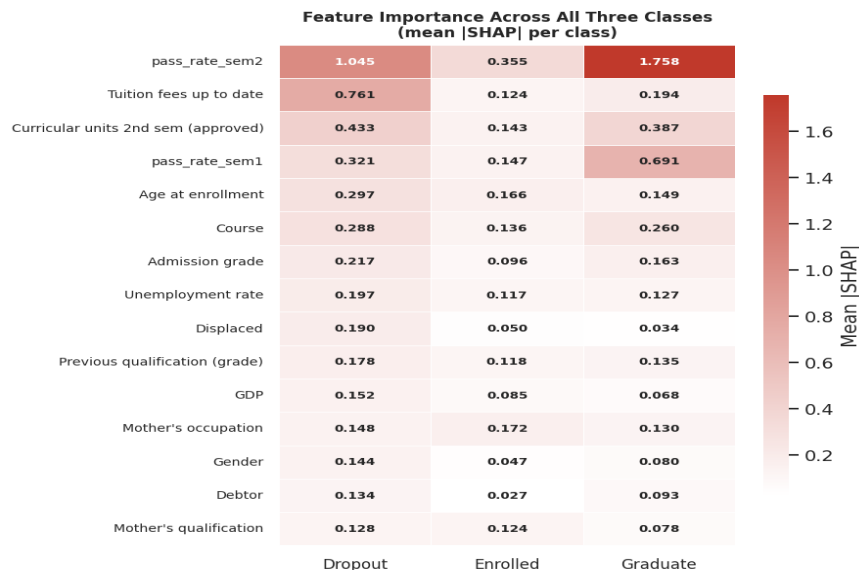
pass_rate_sem2 = 1.045 (37% stronger than next feature)

Semester 2 pass rate is the dominant signal. The standard deviation bar is narrow — this feature is consistently impactful across all test students, not just on average.

Top-2 map to two intervention channels

pass_rate_sem2 (1.045) drives the ACT recommendation. Tuition fees up to date (0.761) drives FIN. The recommender system is grounded directly in these SHAP

Same top features across all three outcome classes (mean |SHAP| per cell)



Heatmap: pass_rate_sem2 is top driver for all three classes

Dropout column 1.045, Enrolled 0.355, Graduate 1.758. The same feature defines both graduation success and dropout risk — just in opposite directions on the prediction scale.

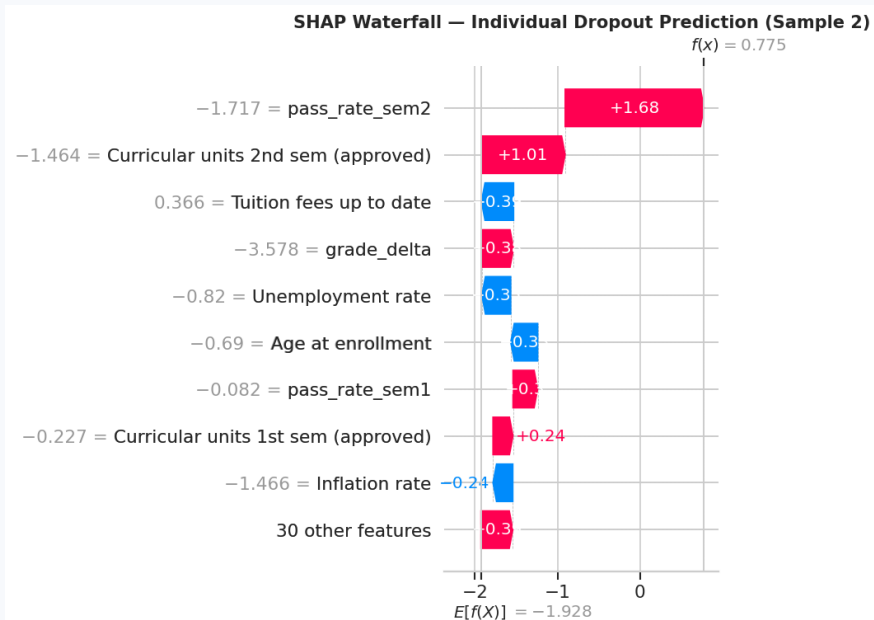
Tuition fees: high for Dropout (0.761), low for Enrolled and Graduate

Financial non-payment is almost exclusively a Dropout signal. Enrolled and Graduate students have comparable low fee-SHAP values, confirming it is a Dropout-specific

SHAP Explainability — Individual Prediction & Key Feature Driver

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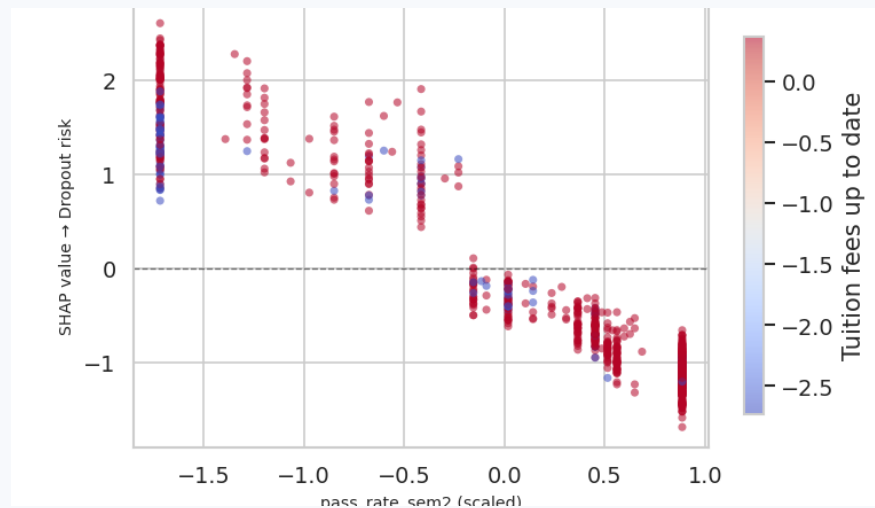
Waterfall Chart — why this specific student was flagged



Reading the waterfall: base 0.32 → final 0.775

Each bar is one feature's contribution to THIS student's prediction. The two dominant red bars — pass_rate_sem2 (+1.68 SHAP) and Curricular units 2nd sem approved (+1.01) — together push the score from the population base rate of 0.32 to 0.775. Tuition fees up to date (0.366) contributes a smaller positive bar. An advisor needs only two words: zero approvals, overdue fees.

Dependence Plot — pass_rate_sem2 (Top #1 SHAP Driver)

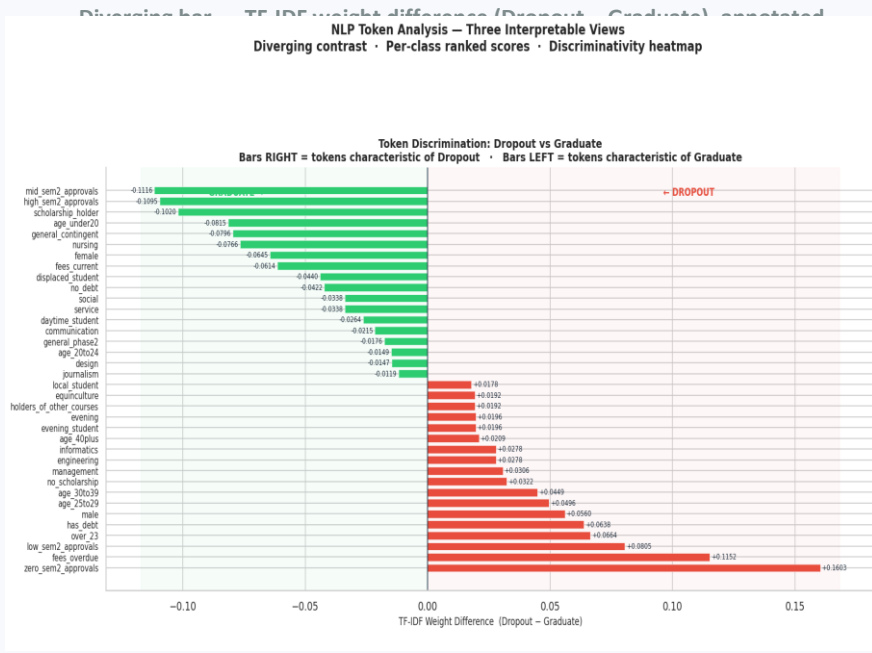


How to read: X = pass_rate_sem2 value · Y = SHAP impact on dropout risk

Left cluster (pass_rate_sem2 ≈ -1.7, i.e. rate ≈ 0): SHAP is +1.5 to +2.5 — these students are the waterfall student's archetype. Moving right as the rate increases, SHAP falls steadily and crosses zero near rate ≈ 0 (the dashed line). Right side (rate > 0.5): SHAP turns negative — a high pass rate actively protects against dropout. Colour = Tuition fees: red dots (overdue) sit higher across all pass rates — compounding the academic risk with financial risk.

NLP Analysis — Token Discrimination: Dropout vs Graduate

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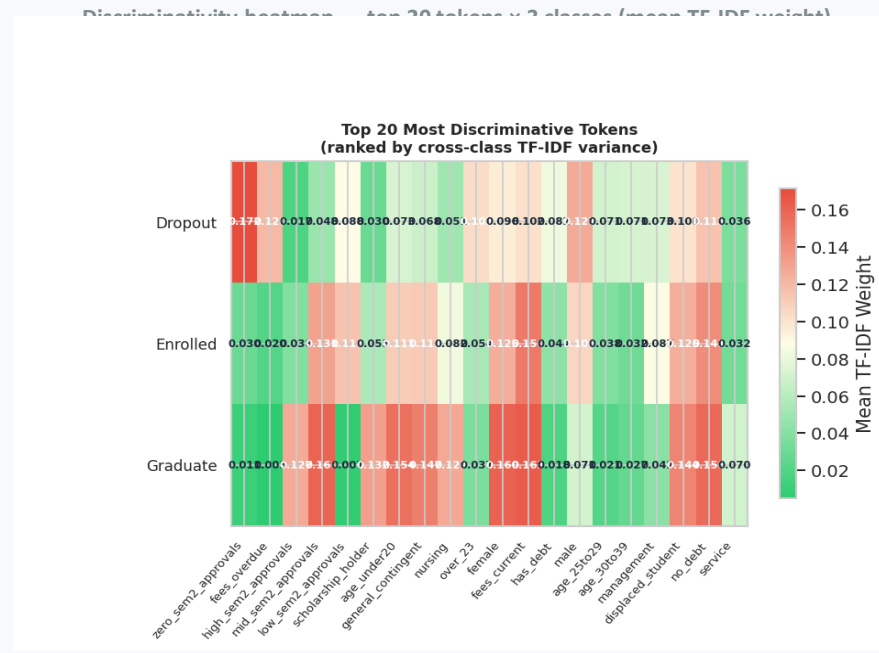


zero_sem2_approvals: largest positive delta (+0.1603)

Appears almost exclusively in Dropout text profiles. fees_overdue (+0.1152) second. Both mirror the SHAP top-2 — entirely different methodology, identical signal. Cross-validates the numerical model.

Graduate tokens: high_sem2_approvals, scholarship_holder, fees_current

The Graduate archetype is young, fee-current, scholarship-holding, and progressing through coursework — exactly the low-risk profile SHAP identifies on the numerical



zero_sem2_approvals scores ~10x higher in Dropout than Graduate

Dark red in Dropout row, near-white in Graduate row. The colour gap quantifies the diverging bar directionally — one token alone provides near-complete class separation.

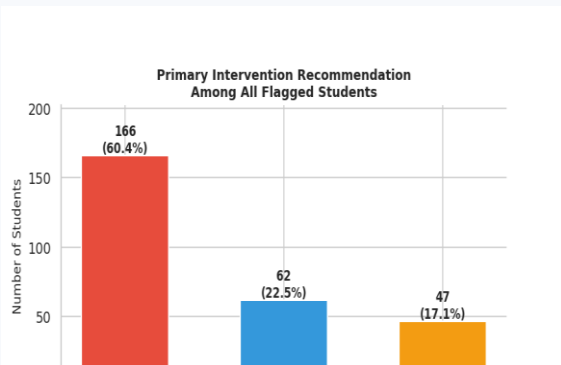
NB text classifier F1 = 0.607 using zero numerical features

Only categorical labels as text. Still 0.385 above the dummy baseline. The 0.088 gap to LightGBM (0.695) precisely quantifies how much semester performance numbers

§ 11 · Intervention Recommender System — SHAP-Weighted Routing

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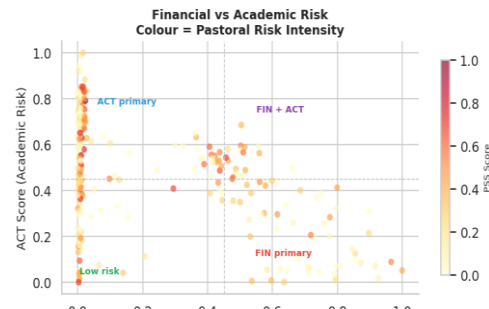
Primary Recommendation Distribution



ACT 166 (60.4%) · FIN 62 (22.5%) · PSS 47 (17.1%)

Academic Support dominates because pass_rate_sem2 is the #1 dropout driver (SHAP 1.045). The split is not uniform — 60% ACT confirms that disengagement, not financial pressure, is the primary dropout mechanism for most students.

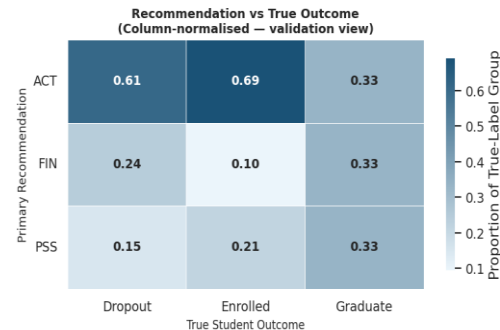
Financial vs Academic Risk — Student Subgroups



Four structural subgroups among flagged students

ACT primary (upper-left): academically disengaged, finances OK. FIN primary (lower-right): fees overdue, grades acceptable. FIN+ACT (upper-right): highest-risk, needs coordinated outreach. Low risk (lower-left): borderline flagged.

Recommendation vs True Outcome — Validation



Validation: 61% of true Dropouts routed to ACT

Among true Dropout students, ACT receives 0.61 — correct primary routing. FIN receives 0.24 for Dropouts. Graduate false-positives split evenly (0.33 each) — low-cost false alarm. Enrolled spreads across all three domains, consistent with their structural ambiguity.

275

Flagged (≥ 0.40)

215 — 78.2%

True Dropouts in Flagged

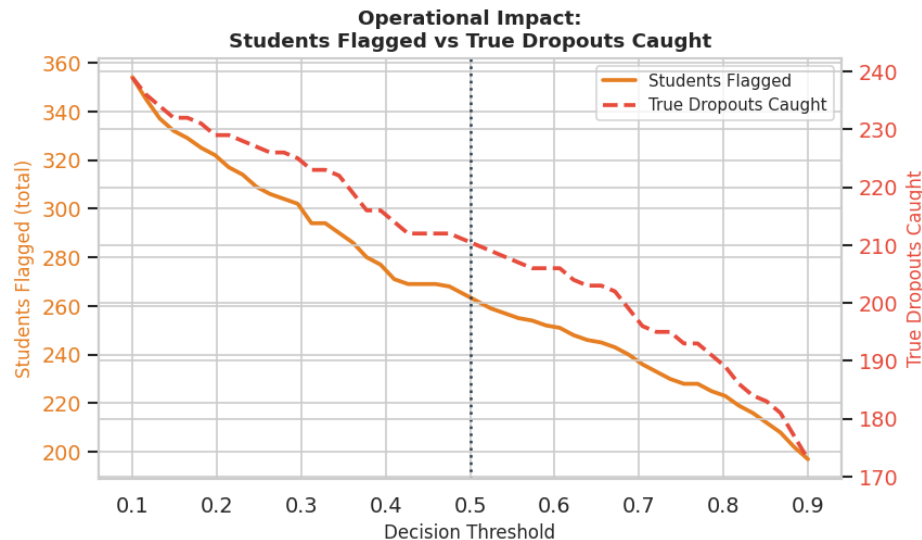
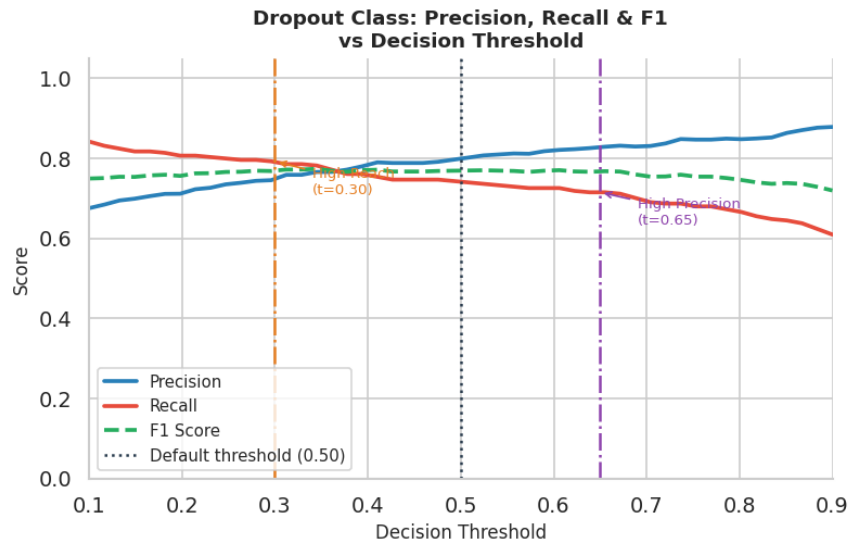
SHAP-weighted content-based

Method

156 — 56.7%

ACT activated (≥ 0.45)

LightGBM (Tuned) — Dropout Threshold Analysis Choosing the Right Operating Point for Institutional Deployment



Default $t = 0.50$

263 students flagged · ~211 true dropouts caught (80.2% precision). Standard balanced operating point — recommended for institutions with moderate counsellor capacity.

High Reach $t = 0.30$

298 students flagged · ~225 true dropouts caught. Higher recall — fewer at-risk students missed. Accepts more false positives. Best when counsellor outreach is low-cost.

High Precision $t = 0.65$

246 students flagged · ~203 true dropouts caught. Fewer false alarms — every flagged student is high-confidence. Best when intervention is resource-intensive (1-on-1 sessions).

Right chart — impact

As threshold rises, students flagged (orange) and true dropouts caught (red dashed) both fall. At $t=0.90$ almost nobody is flagged. Institution should calibrate t against its actual

Evaluation Findings — Best Model Verdict

Why LightGBM wins and what the remaining gaps mean

★ Why LightGBM Won

Criterion	LightGBM	Next Best (RF)
Test Macro F1	0.695	0.708
Macro AUC	0.879	0.887
Dropout Recall	75%	73%
Enrolled Recall	44%	52%
Train/Val Gap	Tight ✓	Wide
Calibration	Best ✓	Moderate
Inference Speed	Fast ✓	Slow

Note on test F1: RF (0.708) has a marginally higher test F1 than LightGBM (0.695), but LightGBM wins on AUC, calibration, and deployment speed — and its tighter learning curve indicates it will generalise better on unseen cohorts.

✓ Strong — Graduate (Recall 88%)

Effectively solved. High pass rates, current fees, younger age create a clean separable cluster. False negatives ($\approx 8\%$) go to Enrolled — never flagged as at-risk.

✓ Deployable — Dropout (Recall 75%)

3-in-4 caught. AUC 0.912 means reliable risk ranking at all thresholds. Missed 25% are borderline cases — probability thresholding and engagement data can recover them.

⚠ Structural Limit — Enrolled (Recall 44%)

Near-random split between Graduate and Dropout errors. Not a model failure — Enrolled is a snapshot of students in motion. Needs longitudinal semester-by-semester data to resolve.

Recommendations & Deployment Roadmap

1 Semesterly Risk Scoring

After Semester 1 closes, score all enrolled students using the saved LightGBM model. Flag anyone with dropout probability ≥ 0.50 for structured advisor outreach. The pipeline takes seconds on a standard institution server.

2 Three-Channel Intervention Routing

Route each flagged student via the SHAP-weighted recommender: FIN $\rightarrow$ financial aid office. ACT $\rightarrow$ academic tutors. PSS $\rightarrow$ welfare/scheduling support. 60.4% will route to ACT — pre-allocate tutor capacity accordingly.

3 Threshold Calibration by Capacity

At $t=0.30$: 298 flagged, ~225 true dropouts caught. At $t=0.50$: 263 flagged, ~211 caught. At $t=0.65$: 246 flagged, ~203 caught. Choose the threshold based on available counsellor hours per cohort.

4 Feedback Loop & Annual Retraining

Log intervention outcomes — who was flagged, who received support, who ultimately stayed or withdrew. Retrain annually on updated cohort data. Add LMS engagement proxies (login frequency, assignment submission) to close the 44% Enrolled ceiling.

Model Limitations (what this system cannot do):

① Cannot predict dropouts for current-semester students without any performance data — features are end-of-semester. ② SHAP gives associations, not causes — a financial intervention may not help a student whose risk is primarily academic. ③ Trained on Portuguese polytechnic data — transfer to other systems requires local retraining. ④ Estimated impact at $t=0.50$: ~211 genuine at-risk students reached per 885-student cohort who would otherwise receive no proactive outreach.